Multi-Disciplinary Minor in Civil Engineering Report

in

Final Year Minor Project -I (CED436)

On

SMART PARKING MANAGEMENT SYSTEM

Submitted by

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Under the Guidance of

Prof. Shrikant Pawar

In partial fulfillment of the requirements for the award of

Bachelor of Technology Multi-Disciplinary Minor in Civil Engineering



Department of Civil Engineering

G.S. Mandal's Maharashtra Institute of Technology Chhatrapati Sambhajinagar

Affiliated to Dr. Babasaheb Ambedkar Marathwada University Chhatrapati Sambhajinagar

Academic Year: 2026–2027

DECLARATION

Multi-Disciplinary Minor

in

Final Year Minor Project -I (CED436)

We hereby declare that minor project report entitled

Minor Project Tittle	Smart Parking Management System
Minor Project Guide	Prof. Shrikant Pawar

is prepared by us as part of our academic project work. The report represents our understanding of the system design and proposed implementation. Wherever external documentation has been used, the relevant sources have been listed in the References section. We understand that academic integrity requires accurate representation of the work and proper acknowledgement of sources.

Place: Chhatrapati Sambhajinagar

Date: 28-Oct-26

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CERTIFICATE

Multi-Disciplinary Minor

in

Final Year Minor Project -I (CED436)

This is to certify that the minor project report entitled

Minor Project Tittle	Smart Parking Management System
Minor Project Guide	Prof. Shrikant Pawar

in partial fulfillment of the requirements for the award of **Bachelor of Technology Multi-Disciplinary Minor in Civil Engineering** at G. S. Mandal's Maharashtra Institute of Technology, Chhatrapati Sambhajinagar.

Student Name	Roll Number	Department
Saiprasad Godge	42008	Computer Science and Engineering
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The report is submitted for academic evaluation.

Place: Chhatrapati Sambhajinagar

Date: 28-Oct-26

Guide – Prof. Shrikant Pawar

HCED – Dr. Manish Dixit

APPROVAL CERTIFICATE

Multi-Disciplinary Minor

in

Final Year Minor Project -I (CED436)



This minor project report entitled

Minor Project Tittle	Smart Parking Management System
Minor Project Guide	Prof. Shrikant Pawar

submitted by the students listed below is presented for evaluation in partial fulfillment of the requirements for the **Bachelor of Technology Multi-Disciplinary Minor in Civil Engineering**.

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ACKNOWLEDGEMENT

Multi-Disciplinary Minor

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We acknowledge the official documentation for Python, Flask, Flask-SQLAlchemy, Flask-Login, Bootstrap, SQLite and Werkzeug consulted while preparing the report.

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ABSTRACT

The issues of traffic management have been skyrocketing with the ever-growing number of personal vehicles and urbanization, hitting the high notes in the commercial complexes, educational institutions, hospitals, and other public places. Motorists with low knowledge spend hours searching for an empty parking spot. Not only does it create unnecessary traffic jams, but it also leads to wastage of fuel consumption and an inefficient parking lot management system. Traditional parking management solutions require manual supervision or hardware solutions such as RFID, infrared sensors, ultrasonic sensors, and license plate recognition cameras.

This project consists of a software focused Smart parking management system, which effectively works without using physical sensors on each parking slot or user. In the existing systems, parking slots status are maintained manually and it is very difficult to handle manually. In our application, the database is used and the parking slots are stored in the database and vehicle entry, parking slots and slot reservation details are saved in the database while vehicle exits and the parking slot is free. When a vehicle comes in, it detects the vacant slots which can be assigned to the vehicle and can decide the slot based on its type, distance from entrance, floor, etc. Car reservation details are saved and then parking information will be generated in digital form after the vehicle is parked. After the vehicle reaches the departure bay, the parking details are removed and the parking lot is opened again. The amount to be paid can be calculated in this system if required.

The project deals with the design of system architecture, database, allocation algorithm, workflow model, and functional testing. The evaluation of this system takes into account the evaluation parameters like precision of allocation, response time, parking searching time, throughput of the system, cost of implementation, etc. The actual values will be obtained after the application is tested and demonstrated. Using this approach, which is not highly reliant on the need for dedicated parking sensors, provides a relatively simple and flexible solution for managing digital parking records. This system may be further extended with mobile app, QR based parking tickets, IoT parking sensors, and automatic parking occupancy detection for this system to be viable in real-world parking facilities.

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ABBREVIATIONS

MDM	Multidisciplinary Minor
IoT	Internet of Things
UI	User Interface
DBMS	Database Management System
DFD	Data Flow Diagram
ER	Entity–Relationship
SQL	Structured Query Language
API	Application Programming Interface
EV	Electric Vehicle
ITS	Intelligent Transportation Systems
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets

CHAPTER 1: INTRODUCTION

1.1 Overview

Parking is an important aspect of the transport system. With more and more vehicles moving into urban areas, educational campuses, business premises and all other public places, there is a need for structured parking spaces. But just providing parking spaces doesn't necessarily mean good parking, as many drivers are often not aware of the space availability or the location of the space.

In a typical parking lot, drivers drive into the parking lot and then look for a space, driving up and down the rows or around the floors of a multi-story building. This process can take extra time, especially during peak hours. Multiple drivers looking for parking at the same time can cause excess movement inside and outside the parking lot.

The objective of a Smart Parking Management System is to get a more organised system through digitised data and rules for allocations. A database is created for the details of each parking space and there is a formal way of saying which parking is free. Rather than visual by just looking around a digital dashboard is used.

A software solution The parking management system we suggest is a software solution. We rely on a centrally run app, as well as a database to store information about parking, handle allocation requests, and store reservation information. Where functionality allows, the system may also include an administrator interface to show parking availability and control parking spaces.

It links the Computer Science & Engineering to a real-world infrastructure problem (Civil Engineering); showing how application development, databases and algorithmic decision-making can aid the management of parking facilities.

1.2 Problem Statement

Conventional parking management techniques rely on manual monitoring, manual observation or simple ticketing systems. These techniques may suffice for small parking lots, but soon grow unmanageable as the number of vehicles and available parking spaces increases.

A major challenge is the availability of information in a user-friendly manner for parking space users about which parking slots are free and free parking slots in other floors and/or areas. As they search for free slots, their vehicle might be occupying a free parking space and another vehicle might be moving around.

One more issue is related to record keeping. If the information concerning the entry/exit of the vehicle or the slot to which it has been assigned is not correctly provided, the recorded availability can become different than the real one.

Parking management directly related to physical occupancy can be achieved using sensor-based parking systems, but the cost of installation and management can be high if there is a requirement for sensors at each parking bay.

Hence, we need an organized parking management system that will keep the digital records of the parking spaces, slots allocation, information of reserved parking lots, and an interface to view the information of parking lot availability.

This issue can be tackled by the introduction of a centralized software based system proposed by the project. It will seek to to keep good records of parking data and enhancement of parking processes but cannot ensure the real parking status.

1.3 Objectives

The primary goals of the Smart Parking Management System are:

1. For the creation of a system which will keep a centralized parking slots information.
2. To show the availability of parking places through a user interface.
3. To recommend suitable parking slots according to configured allocation criteria.
4. To suggest parking spaces based on the configured allocation criteria.
5. In order to keep a record of car entry, reservation and exit.
6. To prevent the allocation of a parking space to more than one active reservation.
7. As an administrative view to monitor and control parking spaces.
8. In order to preserve structured information which will be used as evidence for parking activity monitoring.
9. For creating a framework that is extendable for further parking floors and zones.
10. So we wanted to examine the potential for applying the principles of software engineering in a problem involving parking infrastructure.

1.4 Scope of the Project

The extent of this project is to develop a software-oriented parking management system for arranging the parking areas and managing the basic parking operations.

The proposed system covers the following areas:

- Maintaining digital records of parking spaces.
- Showing recorded parking slots availability.
- Suggesting appropriate free blocks based on set guidelines.
- Recording reservation and vehicle details.
- Handling the checkout procedure and updating parking space files.
- Where available granting administrative access to parking information.
- Keeping records of parking spaces and your reservation activity.

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This system may be expanded to include sites such as educational campuses, office parks, shopping malls, hospitals, and other site with parking facilities.

The current project will cover the software and database components only. Technologies such as physical parking sensors, automatic number plate recognition, barrier gates, payment gateway will not be covered unless they are out of scope that have been already implemented separately.

Due to the recorded nature of this system, the parking records have to be updated correctly for the system to be accurate. Additional hardware, such as sensors or camera systems, may be incorporated into future versions to detect physical occupancy.

1.5 Organization of the Report

This report is organized into five chapters.

Chapter 1: Introduction describes the background of the project, problem statement, its aims and objectives, scope and organization of the report**.**

Chapter 2: Review of Literature: reviews approaches for parking management, technologies relevant to parking management, which may also be applied for the proposed system, their limitations and the need for a system as proposed**.**

Chapter 3: System Analysis and Design The system architecture, technology, database structure, and system actors is explained in Authentication, user roles, parking, recommendation, security, dataflow are also described.

Chapter 4: Results and Discussion contains the state of implementation, testing procedures, Screenshots of the working system, problems faced in testing, benefits and discussion of the results.

Chapter 5: Summary and Conclusions summarizes the project, discusses the conclusions, and suggests what could be done differently for future development.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Existing Approaches to Parking Management

The management of parking spaces has already been transformed from own supervision into computerization and sensor (advanced parking management) equipment. Several methods have been suggested in order to ensure the best parking availability information and vehicle identification.

Size of the parking lot, cost of installation, the extent of infrastructure and automation will be the determining factors behind selecting a specific solution.

2.1.1 Manual Parking Management

The cheapest solution to managing parking is to do it manually. Parking attendants keep track of parking spaces by direct observation and/or manual records, and direct cars to available parking spaces.

This technique consumes fewer technology resources and can work for parking lots on a modest scale. Nevertheless, this approach relies on the human presence of parking attendants to record the state of each parking space accurately. When there are many customers, this can pose a challenge.

2.1.2 RFID-Based Parking Management

RFID system identifies cars or users using tags and readers. An RFID tag linked with a vehicle may be scanned at either the entrance or exit to aid with identification and access.

They can also provide some reduction in the manual entry process and support the vehicle tracking process. However RFID based identification does not automatically confirm that each individual parking bay is physically occupied. Mechanisms may be needed to monitor at the slot level.

2.1.3 Sensor-Based Smart Parking

Sensor-based systems employ devices such as magnetic sensors or ultrasound to determine whether or not a space is occupied.

The information can be transmitted to a controller or a server for displaying on a digital screen. These systems could give more direct information about physical occupancy.

But installing and maintaining sensors in a large parking lot could add to the costs of implementation. The system might also require communications infrastructure, power, and occasional equipment maintenance.

2.1.4 Camera-Based Parking Management

Camera systems apply image processing or computer vision techniques that detect cars or determine the occupancy of parking spots.

Yes, a camera can monitor several bays, if the camera location makes this a possibility. These systems can facilitate automation, but lighting, obstructions, and other factors may impact the efficiency of the system.

Cameras must also take vehicle data-protection into account.

2.1.5 Software-Based Parking Management

The parking information for a software-based system is stored in a database, but can offer features such as checking parking availability, allocating spaces, and registering reservations.

The software-only system is not equipped with a sensing element like the sensor-based option and thus does not need any sensing devices installed on each bay. It might be appropriate for a prototype or in a dedicated property where reliable sources exist for updating parking occupancy.

The main limitation is the recorded status may be inaccurate if entry and exit information isn't properly updated.

Table 2.1: Comparison of Existing Parking Management Approaches

Approach	Advantages	Limitations
Manual management	Simple and relatively inexpensive to establish	Depends on staff and manual records
RFID-based management	Supports vehicle identification and entry records	Does not independently detect every bay's occupancy
Sensor-based management	Can detect vehicle presence at individual bays	Requires hardware installation and maintenance
Camera-based management	Can automate selected monitoring tasks	Depends on image quality, camera placement, and processing
Software-based management	Centralized records and easier configuration	Depends on accurate digital status updates

The project plans to show parking slots management and allocation using software based solution.

2.2 Rules, Standards and Technologies Referenced

The system design approach is inspired by ideas relating to web application development, database design and secure software engineering.

Web Application Development

HTML, CSS, and JavaScript: It is a Web technology which can be used to design a user interface to connect to the parking system.

HTML is used to construct the page's structure CSS is used to style its appearance JavaScript can be used to provide the client-side interaction A backend framework can be used to process requests and implement the allocation rules for the system.

Database Management

A database is necessary to register parking slots, data regarding cars, booking and additional logs.

Such details would be stored in a table, related to each other. The primary key would be used to unique identify each record. The primary key or identifier would be used as a foreign key linking reservations to the parking space.

Correct transaction handling is important in case of a reservation as it includes checking of availability and recording an allocation. Such actions should be executed uniformly, not to produce overlapping records.

Input Validation

Input validation prevents bad data from getting into the system.

For instance, the application could verify that all mandatory information about a car is provided and, that if a ticket ID is submitted, it still exists as an active reservation.

Validation must be server-side validation even if the interface does validation on user input before submitting it.

Security Considerations

The system could store details of registrations and users. Access to records should be limited by user type.

If login is used, the password should be stored with a good password hash method. The database query should be performed safely. The administrative functions should require administrator privileges.

These guidelines influence the design that is being proposed. For the report it should note the particular standards or documentation that have been used as they are only to be included where they have been read.

2.3 Limitations of Existing Systems

Existing parking management approaches have different limitations.

Information about each available space may not be current in manual systems. It depends on visual inspection and recordkeeping.

While RFID systems can detect when a vehicle enters a secure gateway, this does not necessarily imply that the bay is full.

Sensor based systems can enhance occupancy detection but require installation, maintenance and communication from the sensor to the system.

Camera-based methods can automate tasks but image quality, lighting, occlusion and privacy needs should be taken into account.

Software-centric systems make managing information easier, but there is always a risk that it will be incorrect if it is digitally updated. A parked vehicle leaving without being checked out of the system could still show up as an occupied space.

These constraints suggest that parking management is a compromise among cost, automatism, reliability and ease of use.

2.4 Need for the Proposed System

The Smart Parking Management System will be used to bring parking data under a single application.

The system supplies a schema of parking spaces with which to allocate the parking spaces by the rules. It also generates a shared file of reservations and vehicle movement data of the reserves, which facilitates the data analysis.

Since the main features of the prototype are software, a software solution suits so that the system does not require installation of sensors on each individual bay.

The project offers a real-world problem of infrastructure to utilize database design, algorithm design, user interface design and software testing.

The system is not removing all the limitation of the current parking systems. They are only improving the handling of the parking and parking procedure books and allotments, while leaving the physical verification of the parking occupancy as a future potential feature.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Overview and Architecture

The Smart Parking Management System is structured with three major parts: User Interface, Application Logic and Database.

The interface is what the driver/operator uses to interact with the system. It can show parking availability, collect information from the vehicle, and show the assigned slot.

Application logic handles requests, validates input, locates available parking spots, and performs reservation and check-in operations.

The database includes records for parking spaces, reservations, users and transactions where relevant.

This compartmentalization facilitates system maintenance: the interface, business rules, and stored data have their own separate areas.

Main Components

1. User Interface

Parking information and the parking reservation are accessible through the user interface. The interface layout will be designed to give guidance on vacancy information and the total number of parking space allocated.

2. Application Logic

The application logic implements the rules of recommending parking places and also recording the reservation. Also, it checks the correctness of requests and makes a checkout.

3. Database

The parking lot information and reservation details are stored in a database. It helps in keeping a track of the parking lot.

4. Administrator Interface

The administrator interface, if enabled, gives you parking status and management options.

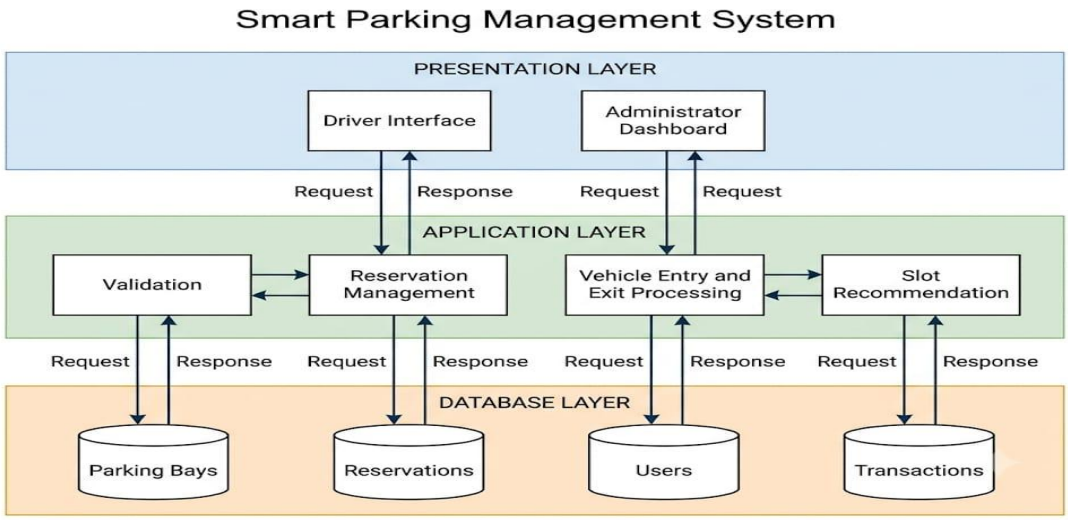


Figure 3.1: Three-tier system architecture diagram showing the user interface, application logic, and database.

3.2 Technology Stack

The stack should list the technologies actually used to build the project.

The following table is a template. If the technology you intend to use is different from the one used as an example, replace the proposed technology.

Table 3.1: Technology Stack

Component	Technology	Purpose
Frontend	HTML, CSS, JavaScript	Creates the user interface
Backend	Python with Flask, if used	Processes requests and implements business logic
Database	SQLite or PostgreSQL, as actually used	Stores parking and reservation information
Database access	SQL or an ORM, if implemented	Reads and updates database records
Version control	Git and GitHub, if used	Tracks source code changes
Deployment	Actual hosting platform or local environment	Runs the application

The frontend is responsible for all interaction and the backend deals with requests and processing them based on known allocation rules. A database is responsible for the storage of parking information.

The technology should be specified with its configuration, copies should be checked on a working project.

3.3 Database Design

One aspect of the proposed system that will benefit from database design is the parking slot allocation as the availability and reservation will need to be stored.

The database is segmented into four tables: parking spaces, reservations, users, and transactions.

3.3.1 Parking Bays Table

This table stores information about each parking space.

Table 3.2: Parking Bays Database Structure

Field	Description
slot_id	Unique identifier of the parking space
floor_level	Floor on which the space is located
zone_name	Section or zone containing the space
bay_type	Type of parking space
status	Recorded availability status
distance_from_gate	Distance or relative position from the entrance
updated_at	Time of the latest status update

The slot identifier provides a unique ID to each parking space. The status field shows the status of the parking space as either recorded as available, reserved, occupied, or marked for maintenance.

3.3.2 Reservations Table

This table stores information about parking reservations.

Table 3.3: Reservations Database Structure

Field	Description
reservation_id	Unique reservation identifier
slot_id	Identifier of the allocated parking space
vehicle_registration	Vehicle registration information
entry_timestamp	Recorded entry or reservation time
exit_timestamp	Recorded exit time
ticket_code	Identifier used to retrieve the reservation
reservation_status	Current state of the reservation

The reservation table links the vehicle information with the assigned parking bay. The it also helps the checkout process by storing the current status of the reservation.

3.3.3 Users Table

A different table could store user information in the case that the user has been authenticated.

Possible fields include:

- User identifier
- Username
- Password hash
- User role
- Account status

Keep sensitive data secure. Only allow administrative access to legitimate accounts.

3.3.4 Transactions Table

If the project includes parking fee calculations, a transaction table can store:

- Transaction identifier
- Reservation identifier
- Parking duration
- Applicable rate
- Calculated amount
- Payment status
- Transaction timestamp

If an actual payment gateway is not incorporated into the application, this should be explained in the report as a fee calculation or presentation of online payment only.

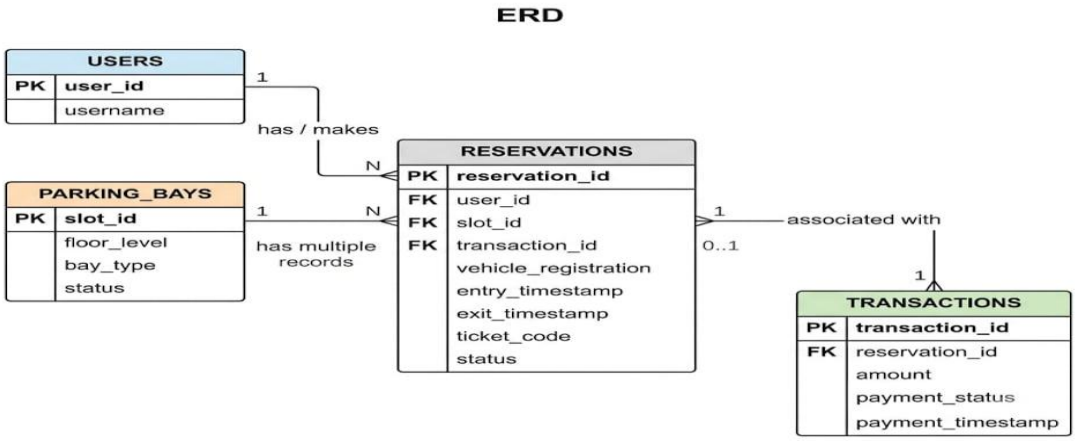


Figure 3.2: Entity-Relationship diagram showing Parking Bays, Reservations, Users and Transactions with their primary keys, foreign keys and relationships.

3.4 User Roles

The system may contain different user roles depending on the implemented features.

Table 3.4: User Roles and Responsibilities

User Role	Responsibilities
Driver	Requests a parking slot and views allocation information
Parking Operator	Assists with entry, checkout, and parking record updates
Administrator	Monitors parking spaces and manages system information

The driver selects a suitable space through the parking interface. An operator may be available to help park where necessary.

The administrator will keep records of the system and will manage how many parking spaces are left. Administration should be limited to authorized persons if access control can be achieved.

3.5 Authentication Flow

When individual user accounts or limited admin functions are present, authentication is required.

A standard authentication workflow starts by the user providing login credentials. The server confirms the validity of the provided credentials and makes sure they match the account record.

Successful authentication results in a session being established, along with access to the system based on the user's role.

If the credentials are not valid, the app prints an error message.

3.6 Core Parking Workflow

The primary workflow involves Slot finding, slot allotting, slot booking, and checkout.

3.6.1 Vehicle Entry and Slot Allocation

This is how it starts, the driver or operator enters the needed information about the vehicle on the app.

The system confirms the data and detects parking slots available for the allocation.

The recommendation logic would then search for the available space that meets the criteria configured, e.g., vehicle type, floor, and nearest to the entrance.

Once an appropriate space is identified, the system verifies the space's availability and then makes the reservation.

The database contains the reservation information. The application will then display the assigned parking space and ticket information.

The driver can then proceed to the designated location.

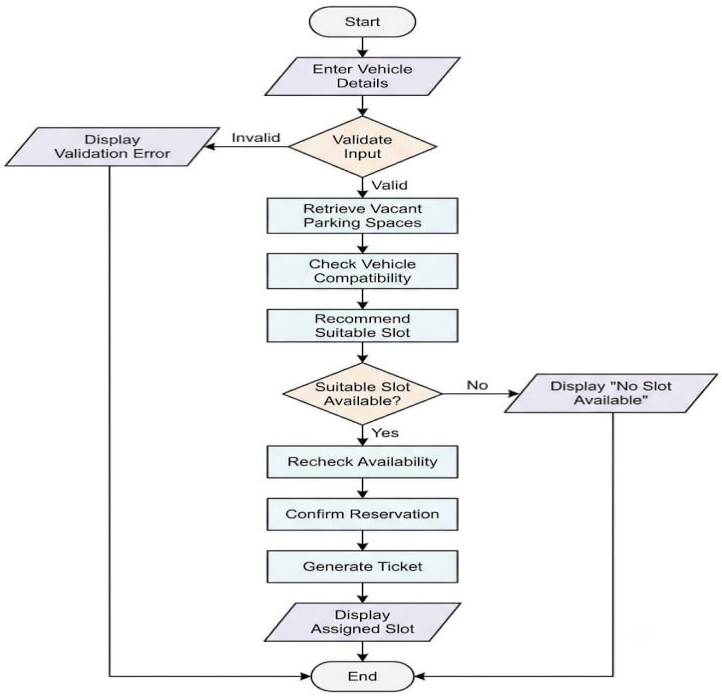


Figure 3.3: Vehicle entry and slot allocation flowchart.

3.6.2 Vehicle Exit and Slot Release

As the vehicle exits the plant, the driver or operator provides the ticket identification.

Whether the ticket is for a reservation that is not yet active.

The exit time is saved if the ticket is correct and the reservation is changed.

In case of applying parking charges, the fee can be generated depending on the length of stay and the applied tariff.

The parking spot can be made available after a successful checkout.

An invalid or already completed ticket should not modify an active parking record.

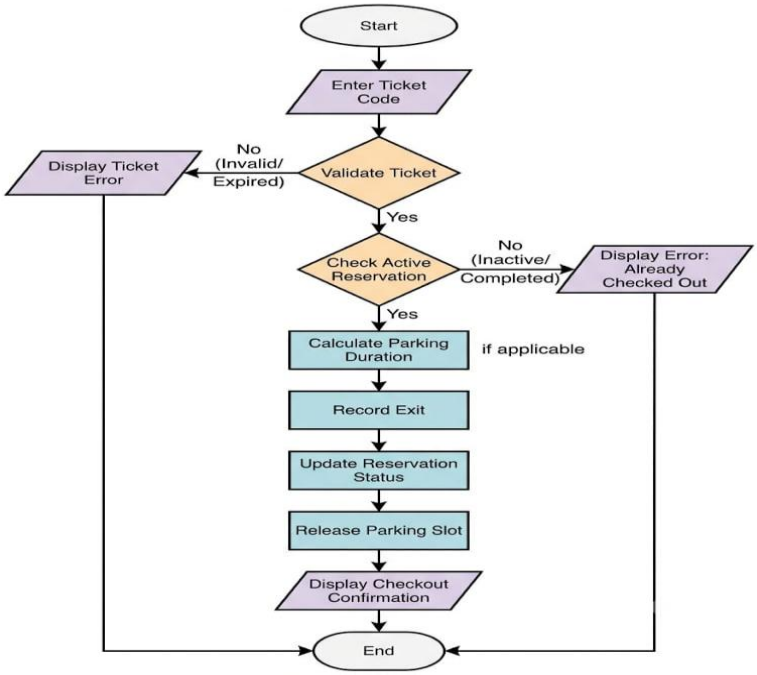


Figure 3.4: Vehicle exit and slot release flowchart.

3.6.3 Parking Slot Status

The system can represent the status of parking spaces using the following states.

Status	Meaning
Vacant	The space is available for allocation
Reserved	The space has been assigned to a reservation
Occupied	The system records a vehicle as occupying the space
Maintenance	The space is temporarily unavailable

The known states should match the deployed application.

For software-only systems, the indication of occupied is based on data logged by application. Do not assume real physical detection. Assuming sensors or other verification will need to be added to make a logical decision.

3.7 Slot Recommendation Process

The slot recommendation process assist you to pick a parking slot from amongst the existing slots.

In lieu of choosing a space at random, the system may be able to consider each of the potential spaces in the context of the established criteria.

The recommendation process may consider:

1. Current recorded availability.
2. Compatibility with the vehicle type.
3. Distance from the entrance.
4. Floor or zone preference.
5. Accessibility requirements, where relevant.

For example, if two parking spaces are vacant and suitable for a vehicle, the system may recommend the one with the lower configured distance score.

A possible scoring formula is:

$$\mathbf{C(S_i) = w_1 \times D(S_i) + w_2 \times F(S_i) + w_3 \times V(S_i)}$$

Where:

- $C(S_i)$ = Total score for parking space S_i.
- $D(S_i)$ = Normalized distance score.
- $F(S_i)$ = Floor preference score.
- $V(S_i)$ = Vehicle compatibility score.
- w_1, w_2, w_3 = Weights assigned to the different criteria.

The weights specify the extent to which each factor affects the recommendation.

It should then be able to select the eligible space which has the lowest score. If there are no spaces which it can chose it should display a message.

If required, the accessibility requirements should be considered as constraints of eligibility, and not just as a regular scoring factor.

3.8 System Notification and User Feedback

The application should provide clear feedback during the main parking operations.

Examples include:

- Verification of receipt of parking request.
- Information about the allocated parking space.
- Message when there is no appropriate space.
- The entered information is incorrect error message.
- Confirmation that checkout has been completed.

Clear messages inform the user whether a request has been successful or not to help the user comprehend what is going on.

Include in this section if there are any features in the project which require email notifications, SMS capabilities, AI assistant etc. and document how it works in reality. These features which are yet to be implemented should be captured in future scope.

3.9 Administrator Workflow

Responsible to the Administrator for keeping parking records and attending the parking lot.

The administrator workflow may include the following activities

1. Accessing the administrative interface.
2. Observing the real-time parking space availability.
3. Reviewing reservation and checkout records.
4. Spaces marked as unavailable when they need maintenance.
5. Restoring spaces to service after maintenance.
6. Checking parking activity reports, if you have reporting enabled.

These functions provide a centralized way to manage parking information.

3.10 Security Design

Because the application will be storing information about vehicle registration, reservations, and user records.

Design-related measures are as follows:

Input validation: The development team should check that the required data has been entered and reject any invalid data**.**

Secure database access: Use of parameterized queries or a reliable database access framework to avoid SQL injection.

Password protection: If authentication takes place, passwords should be stored as hashes, not clear text.

Access control: Administrative functions must be limited to authorized users.

Session management: User sessions should be managed securely, with logout or expiring session behaviour.

Data retention: No personal information, passwords, database credentials, or any other sensitive data should be revealed by the app**.**

But these were the lists of means, not the final result.

3.11 Data Flow Diagram

A Data Flow Diagram (DFD) depicts the way data flows among users, application processes, and data stores.

In the proposed system the driver/operator can request to the slot allocation process. The application can get the available parking spaces from the database. Then it can allocate a parking space.

Once allocated, the system then records the reservation data and informs the user of the ticket data.

The application confirms the ticket during checkout, fetches the reservation, writes the exit record, and clears the space for use by another customer.

The administrator accesses the stored information to monitor parking availability and manage parking records.

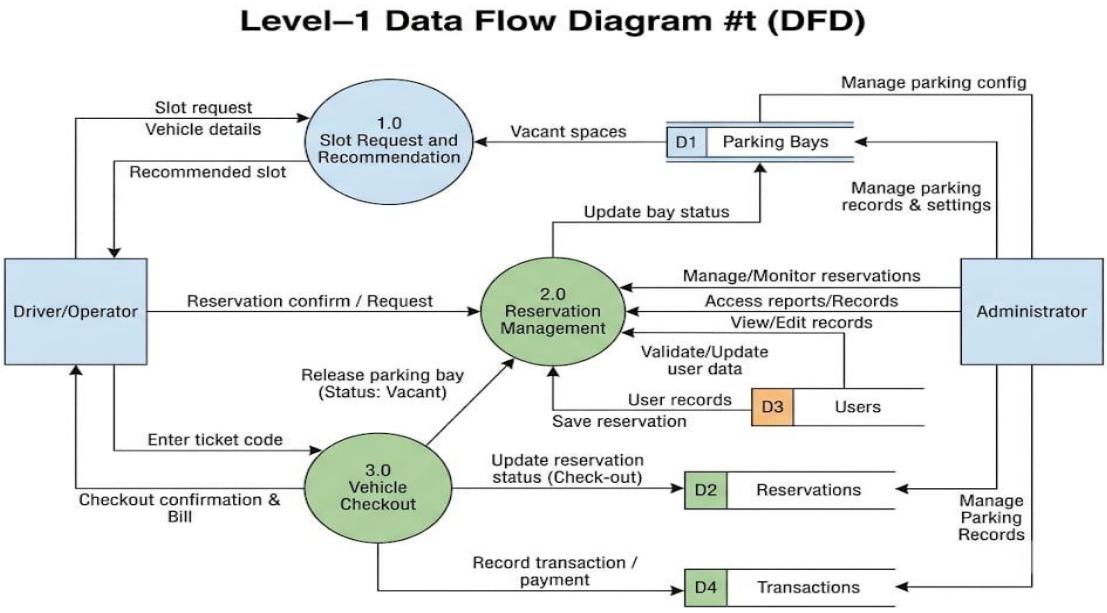


Figure 3.5: Level-1 Data Flow Diagram containing:

- External entities: Driver/Operator and Administrator.
- Processes: Slot Recommendation, Reservation Management, Checkout.
- Data stores: Parking Bays, Reservations, Users, Transactions.
- Flow of data: Slot Request, Availability Details, Details of reservation, ticket details, Checkout details.

3.12 Functional Requirements

The functions specifies the necessity of what the system should do.

Table 3.5: Functional Requirements

ID	Requirement	Description
FR-01	View availability	Display recorded parking slot availability
FR-02	Recommend slot	Identify a suitable vacant space
FR-03	Create reservation	Store reservation and vehicle information
FR-04	Checkout	Validate the ticket and record vehicle exit
FR-05	Manage parking	Allow authorized management of parking records
FR-06	Maintenance status	Exclude unavailable spaces from allocation
FR-07	Fee calculation	Calculate parking charges if implemented
FR-08	Reporting	Display parking activity summaries if implemented

3.13 Non-Functional Requirements

Non-functional requirements describe the expected quality and operational characteristics of the system.

Table 3.6: Non-Functional Requirements

Requirement	Description
Usability	The interface should be understandable and easy to navigate
Reliability	Parking operations should produce consistent records
Data Integrity	A parking space should not have conflicting active allocations
Performance	Common operations should complete within an acceptable time
Security	User information and administrative functions should be protected
Maintainability	The system should be structured for easier modification
Scalability	Additional parking spaces should be manageable without redesigning the entire application

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Implementation Status

This section discusses the progress on implementation.

The implementation should be considered as per modules actually implemented and tested. Design of system gives Parking slot information, allocation, reservation, checkout and admin monitoring.

The table below needs to be filled after verifying the working application.

Table 4.1: Implementation Status

Module	Actual Status	Evidence
User interface	[Complete/Partial/Not Implemented]	Application screenshot
Parking slot records	[Complete/Partial/Not Implemented]	Database or interface
Slot recommendation	[Complete/Partial/Not Implemented]	Allocation output
Reservation management	[Complete/Partial/Not Implemented]	Reservation record
Checkout process	[Complete/Partial/Not Implemented]	Checkout output
Administrator dashboard	[Complete/Partial/Not Implemented]	Dashboard screenshot
Authentication	[Complete/Partial/Not Implemented]	Login test, if applicable
Fee calculation	[Complete/Partial/Not Implemented]	Calculation output, if applicable

The final report should distinguish completed functionality from proposed features.

4.2 Testing

Testing is required to determine whether the application behaves as expected under different conditions.

The test cases should cover normal operations as well as invalid inputs and exceptional situations.

Table 4.2: Test Cases

Test ID	Test Case	Expected Result	Actual Result
TC-01	Request a slot when a suitable space is vacant	The system assigns an eligible space	[Enter result]
TC-02	Request a slot when no eligible space is available	The system displays an appropriate message	[Enter result]
TC-03	Submit incomplete vehicle information	The application rejects invalid input	[Enter result]
TC-04	Attempt to allocate the same remaining space twice	Conflicting active allocations are prevented	[Enter result]
TC-05	Submit a valid ticket for checkout	The reservation is completed and the space is released	[Enter result]
TC-06	Submit an invalid ticket	The system rejects the request	[Enter result]
TC-07	Request a slot when a space is under maintenance	The unavailable space is excluded	[Enter result]
TC-08	Attempt an unauthorized administrative operation	Access is denied, if access control exists	[Enter result]

For each test carried out, the actual results should be noted. If any function has not yet been implemented it should be marked as not applicable/not tested and the reason for this documented.

4.3 Issues Observed During Testing

While developing a parking management application, there are some common problems which we face while managing reservation, updating availability, and checking out.

Below are some prompts to guide this part of the process, but make sure you only list issues that your group experienced.

Issue	Possible Cause	Possible Solution
Incorrect availability status	Parking records were not updated correctly	Validate state changes during reservation and checkout
Duplicate allocation attempt	Availability was not rechecked before confirmation	Use appropriate transaction handling and database constraints
Invalid ticket information	The submitted identifier did not match an active reservation	Validate ticket information before checkout
Incorrect reservation state	A checkout or cancellation operation was incomplete	Review state transitions and update logic
Incorrect recommendation	Eligibility or scoring rules were not applied consistently	Review the allocation criteria and test representative inputs

State how the problem was rectified and what test was conducted to ensure the resolution.

If any issue has not been detected for a specific function, do not invent an issue just to complete the table.

4.4 Advantages of the Proposed System

The following are some of the benefits of our Smart Parking Management System.

1. Centralized Information

Business records of parking space and reservation can be stored centrally, ensuring that the data is easier to find.

2. Systematic Slot Allocation

Instead of making the user choose from the available parking spots, the system can suggest an available parking spot that matches the set conditions.

3. Less Reliance on Bay-Level Detectors

The software-only method, can simulate the primary work flow for the locations, without the need to physically install a sensor at each parking space.

4. Improved Recordkeeping

The barrier to closing the parking journal is reduced with the use of digital reservation and checkout journals.

5. Flexible Configuration

More parking levels and spaces will be able to show in the database records and configuration.

6. Administrative Monitoring

Operators may be able to view availability and block spaces temporarily through an admin interface.

7. Potential for Future Integration

The architecture can be expanded to physical sensors, QR-based tickets, camera systems or other technologies if required.

These benefits are conditional on the system being deployed and used correctly. In particular, a software only system cannot however show whether a bay is physically occupied.

4.5 Discussion

This project on Smart Parking Management System illustrates the application of software and database concepts in parking areas.

A centralized database should be implemented to store information about individual bays and reservations. The suggestion process can be used to suggest a space to the user according to the rules specified in the application.

The tracking system relies on accurately updating the records. For example, if a reservation is made and checked out in the app, but the status is not properly updated, the application can show inaccurate data. Likewise, if a car departs but the checkout is not noted, the space will still be occupied.

These shortcomings bring awareness to the importance of validation, transaction procedures The software-centric architecture should preclude the requirement for custom sensing hardware at each bay during the prototype phase. However, deployment to the real world will likely necessitate physical confirmation, operator processes and additional integration.

Consequently, the outcomes must be considered in the scope of the features used and the tests on the system. Any statements on better search time, accuracy or operational cost have to be documented using measurements or some valid external sources.

CHAPTER 5: SUMMARY AND CONCLUSIONS

5.1 Summary

This project about Smart Parking Management System is based on the issue of managed parking and authenticated parking information.

Our system will utilize an application and database to present the parking slots, keep track of their status, assist with reservation and checkout. Our system will be capable of recommending parking slots by using the configured criteria for allocation.

Additionally, the project takes into account the use of an administrative interface for overseeing parking data and handling vacant spaces.

The report discusses current parking management techniques, such as manual parking, RFID parking management, sensor parking management, camera parking management and software approach. Their advantages and constraints lay the foundation for the design.

The system architecture, database design, user roles, parking process, security issues and testing strategy have been discussed.

The project can show a proof of concept for software based parking coordination and also acknowledge that the available recorded values are only reliable if proper updates are made and additional procedure to monitor the physical occupancy of a parking spot might be needed.

5.2 Conclusions

The Smart Parking Management System is a software-centric system designed to organize parking bays and control reservation data.

Via its centralized recording and predefined allocation scheme the system can contribute to a more structured parking procedure. The reservation process also serves as a foundation for entry/exit recordings and the booking of vacant individual spaces.

This project demonstrates how concepts from computer science including database design and implementation, application development, validation and algorithmic decision making can be used in an infrastructure related project.

The performance of the system will, in part, be limited by the completeness of the records. For example, a solely software-based system will be unable to automatically detect those instances where a vehicle is in an unallocated space or departs without check-out being recorded.

Of course, these limitations can be taken as the basis for enhancement of the system. If implementing in a parking, it would be suitable for actual operation when used in combination with real conditions and supplementary verification or automatical means.

5.3 Possibility of Work in the Future

The system can be upgraded in the following manner.

1. IoT-Based Occupancy Detection

Sensors might also be added to tell us more directly about the actual physical presence of a car in the parking space.

2. Mobile Application

The mobile interface could enable customers to view parking availability and allocations on-the-go as they travel to a site.

3. QR-Based Tickets

QR codes can be utilized for reservation identification. Can be an effective way to validate tickets through the checkout process.

4. Camera-Based Monitoring

Vehicle detection or occupancy estimation could also be investigated using computer vision, assuming accuracy and privacy can be achieved.

5. Parking Analytics

Use historical parking data to analyze the peak times of the day and patterns of using space.

6. Electric Vehicle Parking

The system may be able to cater for a provision of dedicated electric vehicle parking bays and charging facilities.

7. Payment Integration

A payment gateway might also be added to process digital payments after the security and transaction processing measures are in place.

8. Real-World Evaluation

The system could be deployed to a real parking garage to compare the found free spaces with the actual number of cars in the garage and to test the usefulness of the system.

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[5] T. N. Pham, M. F. Tsai, D. B. Nguyen, C. R. Dow, and D. J. Deng, "A Cloud-Based Smart Parking System Using IoT Technology". Make sure to check out the page number and volume.

[6] The actual code for the programming language, backend framework, and database used is pasted here.

[7] List here any other research papers that your group has read and provided input for on the project.